Assignment 1

📌 Assignment Instructions

📢 Important Notice

This assignment covers **key concepts** in Natural Language Processing (NLP), including:
✅ **Regular Expressions (Regex)** for pattern matching
✅ **Minimum Edit Distance (Levenshtein Distance)** for text similarity and error correction
✅ **N-Grams** for predictive text and language modeling

📝 Submission Guidelines

- **All answers must be handwritten** and submitted using the answer sheet provided at the following link**.**
- Download and print the answer sheet and write your answers manually on the answer sheet.
- Scan all the pages in the correct sequence, create a PDF document and submit to eclass.
- Please, also bring the actual paper submission to the first class after submission and submit it to the professor.
- Ensure your handwriting is **clear and legible**.
- If your handwriting is **not readable**, **you will not receive a score for that task**.

🔹 Formatting & Presentation

- **Write neatly and in an organized manner.**
- Clearly label each task and **show all necessary steps**.
- **Illegible or incomplete responses will not be graded.**

📅 Deadline

- **Submission due by: 29.03.2026 Sunday [23:55]**
- **Late submissions:**
  - **within 24 hours: 50% penalty**
  - **Beyond 24 hours: not accepted**

⚠️ Academic Integrity

- **This is an individual assignment.**
- **Copying from others or AI-generated solutions is strictly prohibited.**
- Any form of plagiarism will result in a **zero score** for the assignment.

Task 1. Regex

Provide **one example** for each of the following regular expressions while **complying with the following conditions** and also provide explanation of each part and the corresponding value you provided(see my example below later).
Conditions:
When you need to use a letter, you should use the letters in the reverse order of your name(also uzbek national suffixes(~bek, jon, khon, etc.) removed from your name for uniqueness).
When you need to use a number, you should use the numbers in the reverse order of the last 4 digits of your ID number.
Not following the above rule will result in 0 score.

Example:
Student name: Lazizbek, ID: U1810021
Reversed name: zizaL(bek removed; these are the letters you can use and have to use in the current order, you can change the letter from upper to lowercase or vice versa),
Last 4 digits of ID reversed: 1200

Sample regex:
`^[A-Z][a-z]+[0-9]{3,5}-[A-Z]{2}$`
Breakdown of components:
`^[A-Z]`-> Start with an uppercase letter -> Z (first letter I have in my reversed name is ‘z’ and I made it uppercase to match)
`[a-z]+` -> at least 1 lowercase letter -> iz (I took 2 letters after the first ‘z’ that I used above)
`[0-9]{3,5}` -> 3 to 5 digits -> 1200 -> I took all 4 of reversed last 4 digits of my ID
`-[A-Z]{2}$` -> ends with a dash and 2 uppercase letters -> -AL

Final result: Ziz1200-AL

See next page for sample written submission.

STUDENT NAME: Lazizbek Kahramonov

STUDENT ID NUMBER: 4 181 0021

SECTION NUMBER IN THIS COURSE: 001

Reversed name: zizaL (suffix removed)

Last 4 digits of ID reversed: 1200

Regex 1. ^[A-Z][a-z]+[0-9]{3-5}-[A-Z]{2}$

Breakdown of components

1) ^[A-Z] → start with an uppercase → Z ← zizaL

2) [a-z]+ → at least 1 lowercase letter → iz ← zizaL

3) [0-9]{3,5} → 3to5 digits → 1200 ← 1200

4) -[A-Z]{2}$ → ends with "-" and 2 uppercase letters → AL ← zizaL

Final result: $\boxed{\text{Ziz1200-AL}}$

Task 2. Minimum Edit Distance Calculation.

Minimum edit distance algorithm is used in spell-checkers. In this task we will apply this using your names.

Calculate the minimum edit distance between your name (as the source string) and 2 of your friends' names (as the target strings one by one), all made lowercase and uzbek national suffixes(~bek, jon, khon, etc.) removed for simplicity.

For example, my name is Lazizbek and my friends' names are Behruz and Nodir.
The target strings in this case would be "behruz" and "nodir" while the source string is "laziz".

Please follow this video and provide submission as shown in the video and on the **next page**, clearly drawing the arrows indicating the source cell of the value(which cell was added +1 to generate the target cell value).

- Right arrow indicates "Insert"
- Down Arrow indicates "Remove|Delete"
- Diagonal Arrow indicates
    - "Copy"(that is to just keep) if characters being compared are SAME
    - "Replace" otherwise

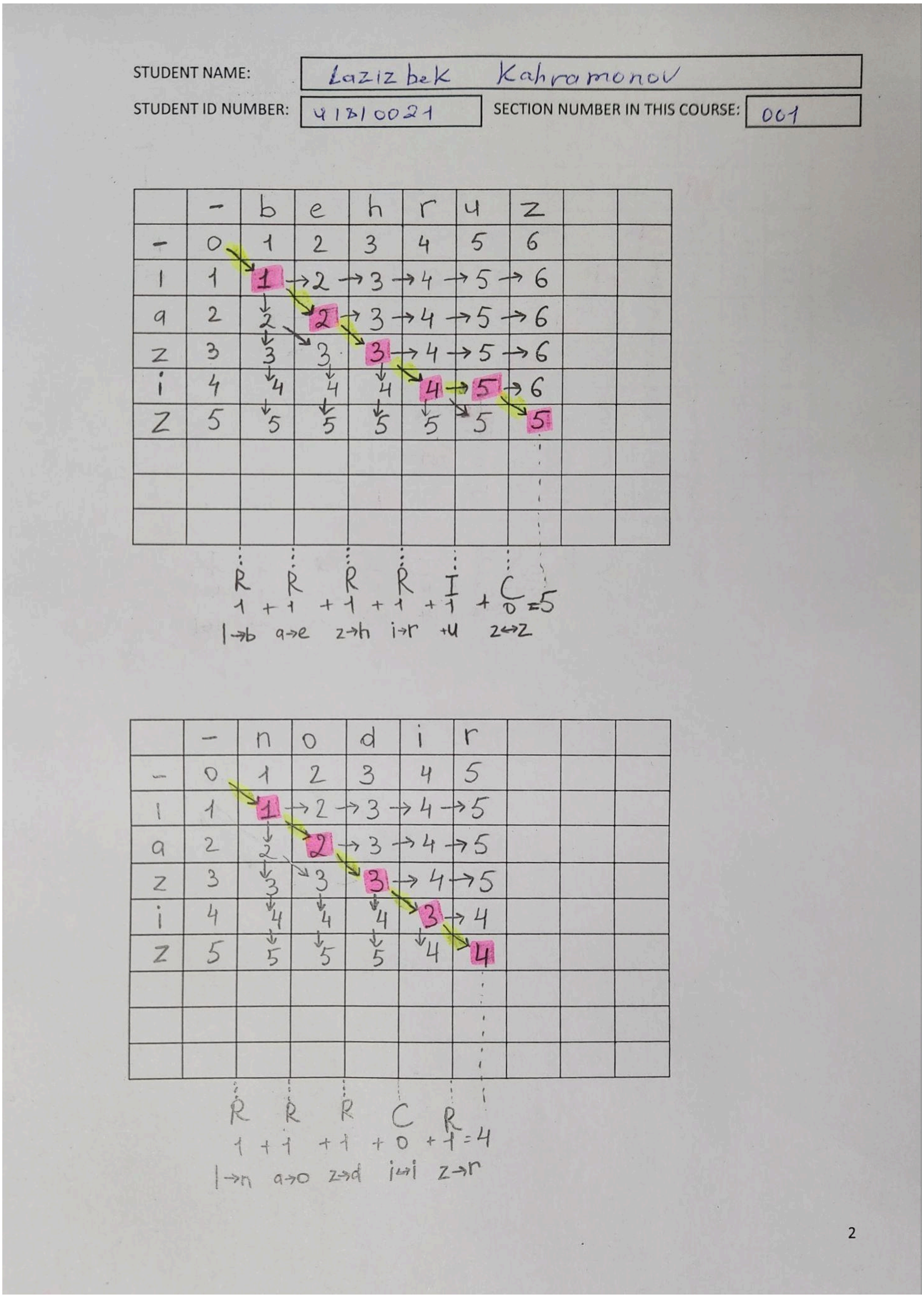
Cost of all operations - Insert, Remove, Replace - is 1.

First, find edit distance with your first friend and in the next table with your second friend and compare whose name is closer to yours.

IMPORTANT
Use the following capital letters to indicate the action taken below each line corresponding to the arrow just like I showed.
I: Insert, **D**: Delete, **R**: Replace, **C**: Copy(Keep)
Those letters should be directly under the vertical line of the table column where the corresponding arrow is drawn above.(see my dotted lines below to indicate that)
Underneath each operation, write down costs and sum up to find the final cost.
Directly underneath each operation, also write down which letters participated in that operation(l -> b, meaning 'l' was replaced with 'b' in the below example.)

Your submission should look like mine except for the names. Use your own name and 2 different friends' names. I provided 10x10 tables, you can expand them or just include the names until the max capacity of the table.

Find a sample submission using my name on the next page.



STUDENT NAME: Lazizbek Kahramonov

STUDENT ID NUMBER: 41810021 SECTION NUMBER IN THIS COURSE: 001

	-	b	e	h	r	u	z		
-	0	1	2	3	4	5	6		
l	1	1	2	3	4	5	6		
a	2	2	2	3	4	5	6		
z	3	3	3	3	4	5	6		
i	4	4	4	4	4	5	6		
z	5	5	5	5	5	5	5		

R R R R I C

1 + 1 + 1 + 1 + 1 + 0 = 5

l→b a→e z→h i→r +u z↔z

	-	n	o	d	i	r			
-	0	1	2	3	4	5			
l	1	1	2	3	4	5			
a	2	2	2	3	4	5			
z	3	3	3	3	4	5			
i	4	4	4	4	3	4			
z	5	5	5	5	4	4			

R R R C R

1 + 1 + 1 + 0 + 1 = 4

l→n a→o z→d i↔i z→r

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It is important to first watch the video above to understand how it works and then apply here.
Here are references for more related content

- Leetcode video, Edit Distance Between 2 Strings - The Levenshtein Distance ("…

Task 3. N-Grams for Predictive Text & Language Modeling

As you know, Smartphones and AI assistants use N-Gram models to predict words as we type.
So, in this task, you will also try to implement a predictive typing system.

Given this user input: “machine learning is"

1. Construct bigrams from the following training sentences:
   a. "machine learning is powerful"
   b. "machine learning is difficult"
   c. "machine learning is useful"
   d. "machine learning is fun"
2. Compute **bigram probabilities** using:

$$P(w_n|w_{n-1}) = \frac{\text{Count}(w_{n-1}w_n)}{\text{Count}(w_{n-1})}$$

3. Predict the next word using the highest probability.

Please use and fill in the answer sheet corresponding sections to address each of the above sub-tasks.